

Imputation continued

Why impute?

Some algorithms do not run with NA's.

Some algorithms default to dropping NA's (listwise deletion).

Imputing lets you decide for yourself.

Why data is missing

MCAR (Missing completely at random):

Which entry gets erased from the complete dataset is randomly selected, without knowing the dataset itself.

MAR (Missing at random):

Probability that an entry is missing may depend on observed data, but not on the values missing.

MNAR (Missing not at random): This case is the negation of MAR.

Taken from:

https://www.youtube.com/playlist?list=PL6tc6IBlZmOV0LRP_Hihmc0D-JwF6viA4

Lessons by Mikko Rönkkö on Sampling, sample selection, and missing data

TABLE 2.1. Employee Selection Data Set

Complete data		Missing data
	Job performance	Job Performance
IQ		
78	9	—
84	13	—
84	10	—
85	8	—
87	7	—
91	7	—
92	9	—
94	9	—
94	11	—
96	7	—
99	7	7
105	10	10
105	11	11
106	15	15
108	10	10
112	10	10
113	12	12
115	14	14

Algorithm can yield

Consistent estimators:

Estimator converges to the parameter as sample size grows larger and larger.

Efficient estimators:

Can already get good estimation with smaller dataset compared to other methods.

Missing data mechanism	List-wise deletion	Mean imputation	Modern missing data techniques	Selection models
MCAR	Consistent Inefficient	Inconsistent Inefficient	Consistent Less Inefficient	Not applicable
MAR	Inconsistent Inefficient	Inconsistent Inefficient	Consistent Less Inefficient	Consistent, strong assumptions
MNAR	Inconsistent Inefficient	Inconsistent Inefficient	Inconsistent Less Inefficient	Consistent, strong assumptions

Learn more:

https://www.youtube.com/playlist?list=PL6tc6IBIZmOV0LRP_Hihmc0D-JwF6viA4

Lessons by Mikko Rönkkö on Sampling, sample selection, and missing data

Why impute? Some algorithms do not run with missing values by default before you choose how to prepare your data.

Types of missing data:

MCAR

MAR

MNAR

The algorithm can yield

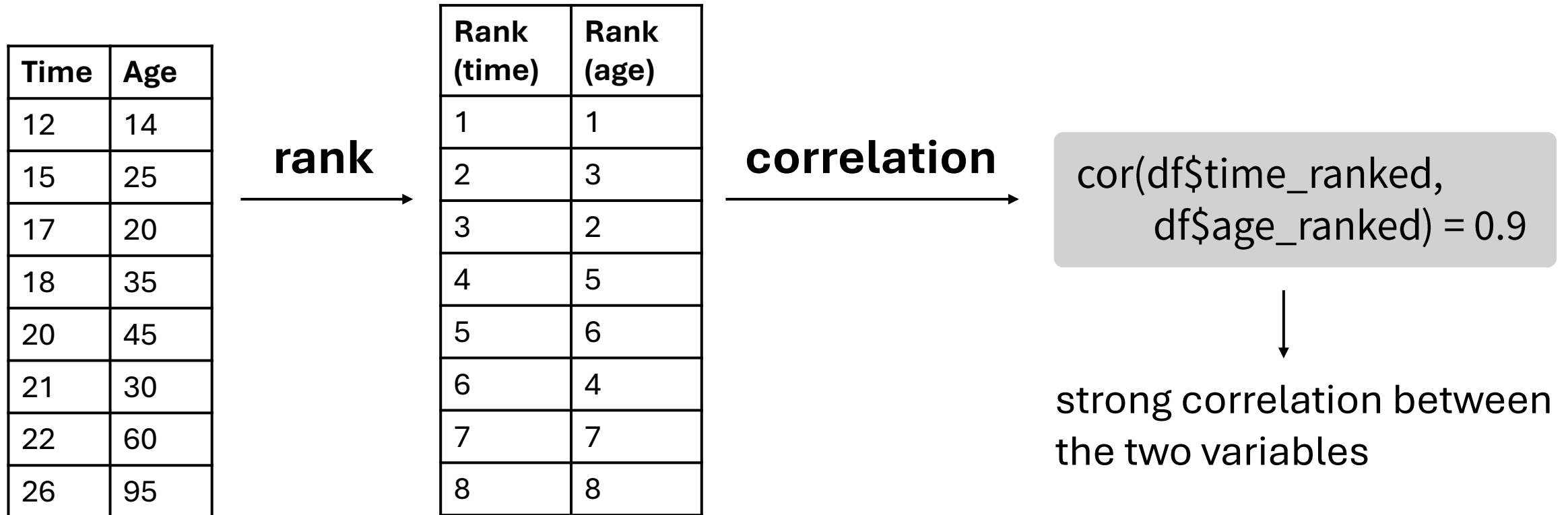
- Consistent estimators i.e. as sample size grows large, estimator gets close to the parameter.
- Efficient estimators i.e. good estimates with small dataset relative to other methods.

Theoretical data type conversions

Conversions: ordinal data

Encode **ordinal data** by its **rank**. Correlation here is known as **Spearman rank correlation** ρ and measures non-parametric relationship between two variables.

Example (numiqo). Gamer reaction time related to age?



Conversions: categorical data

Encode **categorical data** by a series of **Yes/No questions**.
This is called **one-hot encoding**.

Name	Lunch
Alisha	Pizza
Alex	Apple
Bella	Tea
Carter	Pizza
Colin	Pizza
Jacob	Tea
Joshua	Tea

one-hot

Name	Apple?	Pizza?	Tea?
Alisha	0	1	0
Alex	1	0	0
Bella	0	0	1
Carter	0	1	0
Colin	0	1	0
Jacob	0	0	1
Joshua	0	0	1

```
df$apple <- (df$lunch == "apple")  
df$pizza <- (df$lunch == "pizza")  
df$tea <- (df$lunch == "tea")
```

Conversions: categorical data

Encode **categorical data** by a series of **Yes/No questions**.
This is called **one-hot encoding**.

Name	Sick?
Alisha	Yes
Alex	No
Bella	Yes
Carter	No
Colin	Yes
Jacob	No
Joshua	No

Conversions: categorical data

Encode **categorical data** by a series of **Yes/No questions**.
This is called **one-hot encoding**.

Name	Sick?
Alisha	Yes
Alex	No
Bella	Yes
Carter	No
Colin	Yes
Jacob	No
Joshua	No

one-hot



Name	Yes to Sick?	No to Sick?
Alisha	1	0
Alex	0	1
Bella	1	0
Carter	0	1
Colin	1	0
Jacob	0	1
Joshua	0	1

Conversions: categorical data

Encode **categorical data** by a series of **Yes/No questions**.
This is called **one-hot encoding**.

Name	Sick?
Alisha	Yes
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Bella	Yes
Carter	No
Colin	Yes
Jacob	No
Joshua	No

↔
Dummy variable encoding

one-hot



Name	Yes to Sick?	No to Sick?
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Alex	0	1
Bella	1	0
Carter	0	1
Colin	1	0
Jacob	0	1
Joshua	0	1

↔
Dummy variable encoding

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Dummy variable encoding

one-hot



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Dummy variable encoding

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one-hot

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Alex	1	0	0
Bella	0	0	1
Carter	0	1	0
Colin	0	1	0
Jacob	0	0	1
Joshua	0	0	1

```
df$apple <- (df$lunch == "apple")  
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df$tea <- (df$lunch == "tea")
```

Dummy variable encoding uses one less variable than one-hot encoding

Conversions: categorical data

Encode **categorical data** by a series of **Yes/No questions**.
This is called **one-hot encoding**.

Name	Lunch
Alisha	Pizza
Alex	Apple
Bella	Tea
Carter	Pizza
Colin	Pizza
Jacob	Tea
Joshua	Tea

one-hot

Name	Apple?	Pizza?	Tea?
Alisha	0	1	0
Alex	1	0	0
Bella	0	0	1
Carter	0	1	0
Colin	0	1	0
Jacob	0	0	1
Joshua	0	0	1

```
df$apple <- (df$lunch == "apple")  
df$pizza <- (df$lunch == "pizza")  
df$tea <- (df$lunch == "tea")
```

Dummy variable encoding omits a variable from one-hot encoding

Review for Exam 1

Get in groups to do practice exam.